ELECTRIC VEHICLE MARKET PERFORMANCE & INSIGHTS

PROJECT REPORT

Data Analysis and Visualization using Tableau

GitHub Repository

github.com/khuship9575/EV-Market-Performance-Insights

PROJECT REPORT

Electric Vehicle Market Performance & Insights

1. Introduction

The Electric Vehicle Market Performance & Insights project analyzes electric vehicle market data using Tableau. It presents information about manufacturers, sales, market regions, and battery technologies through charts and an interactive dashboard.

2. Objectives

- Analyze electric vehicle market data.
- Identify top-performing EV manufacturers.
- Compare EV sales across market regions.
- Understand battery technology distribution.
- Create an interactive dashboard for analysis.

3. Dataset Description

The project uses the **ev_market_train.csv** dataset containing 200,000 records and 37 columns. It includes manufacturer, market region, annual sales units, vehicle category, battery type, vehicle range, price, and customer rating.

4. Tools and Technologies

- Tableau – data visualization and dashboard creation.
- CSV dataset – data storage and analysis.
- GitHub – project file management and documentation.

5. Methodology

- Collect the EV market dataset.
- Prepare data and check missing values.
- Connect the dataset to Tableau.
- Create charts for sales and battery distribution.
- Develop an interactive dashboard.
- Publish the dashboard on Tableau Public.
- Upload project files and documentation to GitHub.

6. Visualizations

- Top 10 Manufacturers by Sales.
- EV Sales by Market Region.

- Battery Technology Distribution.

7. Key Findings

- AmpereWorks recorded the highest manufacturer sales total in the dataset.
- South Asia recorded the highest regional sales total.
- NMC was the most common battery type by vehicle records.

8. Conclusion

The project demonstrates how Tableau can analyze EV market data and communicate patterns through visual dashboards. The dashboard makes comparisons easier and supports data-driven understanding. Findings are limited to the supplied dataset and are not verified real-world market statistics

9. Project Links

- GitHub: https://github.com/khuship9575/EV-Market-Performance-Insights
- Tableau: https://public.tableau.com/app/profile/vishesh.malpani/viz/EVMARKETPERFORMANCEANDINSIGHTS/ElectricVehicleMarketPerfomanceInsights_1?publish=yes
- Website: https://ev-market-performance-insights-l1ta.onrender.com/